Chapter 14 — Concurrency I: Threads, Locks and the `java.util.concurrent` Toolkit

A web server handling thousands of requests, an app downloading images while the user scrolls, a batch job using all 16 cores: all of these do several things **at the same time**. Java has supported this since version 1.0 through **threads**, and since Java 5 through the high-level `java.util.concurrent` package (designed by Doug Lea and the authors of *Java Concurrency in Practice*, "JCIP"). Concurrency is also where Java programs get their nastiest bugs — results that are wrong only sometimes, on some machines, under load.

This chapter builds the foundation: creating and controlling threads, *why* shared data goes wrong (race conditions and visibility), the tools that make it right (`synchronized`, `volatile`, locks, atomics), and the everyday toolkit you should actually use at work (thread pools, concurrent collections, synchronizers). Chapter 15 continues with `CompletableFuture`, Fork/Join, **virtual threads**, structured concurrency, scoped values, the Java Memory Model and liveness problems (deadlock, livelock, starvation).

What you'll learn

§	Topic	You'll be able to…
14.1	Concurrency basics	explain processes, threads, concurrency vs parallelism, and why threads share memory
14.2	Creating and starting threads	use `Thread`, `Runnable`, lambdas and `Thread.Builder`; `start` vs `run`; `join`; daemon threads
14.3	Thread lifecycle	name every `Thread.State` and what causes each
14.4	Interruption	cancel threads cooperatively and handle `InterruptedException` correctly
14.5	Race conditions	recognise read-modify-write and check-then-act bugs
14.6	`synchronized` and intrinsic locks	protect shared state; know what object is locked
14.7	`volatile` and visibility	stop a thread with a flag; know what `volatile` does *not* do
14.8	`wait` / `notify`	write a guarded block correctly (and know why you rarely should)
14.9	Explicit locks	use `ReentrantLock`, `tryLock`, `Condition`, `ReadWriteLock`, `StampedLock`
14.10	Atomic variables	use `AtomicInteger`, CAS, `AtomicReference`, `LongAdder`
14.11	Executors and thread pools	submit tasks, use `Future`, size and configure pools, shut them down
14.12	Concurrent collections	pick `ConcurrentHashMap`, `CopyOnWriteArrayList`, `BlockingQueue` … and avoid compound-action bugs
14.13	Synchronizers	coordinate threads with `CountDownLatch`, `CyclicBarrier`, `Semaphore`, `Phaser`
14.14	Designing thread-safe classes	apply JCIP's strategies: immutability, confinement, encapsulated locking

All runnable code is in `code/` (package `ch14.*`), compiled with `javac --release 25` and run on JDK 26 on a 14-core laptop. **Concurrency output is not always repeatable.** Lines marked `[nondeterministic]` change from run to run (that is the point being demonstrated); lines marked `[machine-dependent]` depend on your hardware. Everything else is forced into a fixed order with `join`, latches or barriers.

> **Related material — linked, not repeated.**
>
> - Exam-level API tables and trick questions (`Future` methods, `Executors` factories, `TimeUnit`, lock methods, parallel streams): OCP Chapter 13 — Concurrency.
> - Design-level concepts with diagrams and interview framing (mutex, semaphores, producer-consumer, reader-writer, thread-pool patterns): LLD Chapter 12 — Concurrency concepts.

- Classic problems solved step by step (FooBar alternation, H2O, dining philosophers, blocking queue, TTL cache, rate limiter): LLD Chapter 13 – Concurrency problems.

14.1 Concurrency basics

What it is

- A **process** is a running program with its own memory, isolated from other processes by the operating system. Starting `java App` creates one process: the JVM.
- A **thread** is an independent path of execution *inside* a process. Each thread has its own **call stack** (local variables, method frames) and program counter, but **all threads in a process share the same heap** – the same objects and fields.
- **Concurrency** means several tasks are *in progress* during the same period (possibly taking turns on one CPU core). **Parallelism** means tasks literally run *at the same instant* on different cores. Concurrency is about structure; parallelism is about hardware. (Detailed comparison with diagrams: LLD §12.1–12.2.)

Analogy: a restaurant kitchen. The kitchen (process) has shared fridges and counters (heap). Each cook (thread) has their own recipe card and hands (stack). Cooks working at the same time get a lot done – until two of them grab the last egg at once.

Why it matters

1. **Responsiveness** – a UI or server keeps answering while slow work (disk, network) happens elsewhere.
2. **Throughput** – while one request waits for a database, another can use the CPU.
3. **Using all cores** – CPU-heavy work (image processing, sorting big data) can be split across cores.

The core problem

Because threads share the heap, two threads touching the same object can interfere. Three things can go wrong:

1. **Atomicity** – an operation that looks like one step (`count++`) is really several, and threads interleave in the middle (§14.5).
2. **Visibility** – a value written by one thread may not be seen by another, because of CPU caches and compiler reordering (§14.7, and the Java Memory Model in Chapter 15).
3. **Ordering** – operations may appear to happen in a different order to another thread.

Everything in this chapter is a tool against one or more of these.

Key terms

Term	Meaning
Thread-safe	behaves correctly when used by several threads at once, with no extra coordination by callers
Shared mutable state	data that more than one thread can access *and* that can change – the source of all concurrency bugs
Critical section	code that touches shared mutable state and must not be executed by two threads at once
Lock / monitor	a mechanism that lets only one thread (or a controlled number) into a critical section
Blocking	a thread waiting (for a lock, I/O, a sleep) without doing work
Context switch	the OS pausing one thread and running another on the same core

---

14.2 Creating and starting threads

What it is

A `java.lang.Thread` object represents one thread. The *work* it does is a `Runnable` – an interface with one method, `void run()`. Since Java 21 there are two kinds of threads: **platform threads** (a thin wrapper over an OS thread – this chapter) and **virtual threads** (lightweight threads managed by the JVM – Chapter 15 §15.4).

How it works

1. Describe the work: subclass `Thread` and override `run()`, or (better) write a `Runnable` – often a lambda.
2. Create the thread: `new Thread(runnable, "name")`, or the builder `Thread.ofPlatform().name("x").unstarted(runnable)` (Java 21).
3. Call `start()`: the JVM creates a new OS thread, which calls `run()`. `start()` returns immediately; the new thread runs concurrently.
4. Optionally `join()`: the calling thread waits until the other thread has finished.
5. The thread ends when `run()` returns (or throws). A thread can be started **only once**.

`code/ch14/threads/CreatingThreads.java` (key parts):

```
static class Greeter extends Thread {                       // 1) subclass Thread
    Greeter() { super("greeter"); }
    @Override public void run() { System.out.println("1 subclass  running in " +
Thread.currentThread().getName()); }
}
static class Task implements Runnable { ... }               // 2) implement Runnable

Thread t1 = new Greeter();
Thread t2 = new Thread(new Task(), "worker-2");
Thread t3 = new Thread(() -> System.out.println(...), "worker-3");          // 3) lambda
Thread t4 = Thread.ofPlatform().name("worker-4").unstarted(() -> ...);      // 4) builder (Java 21)
for (Thread t : new Thread[]{t1, t2, t3, t4}) { t.start(); t.join(); }

Thread notStarted = new Thread(() -> ...);
notStarted.run();                                           // just a method call - runs in main!
t1.start();                                                 // second start -> IllegalThreadStateException

Thread a = new Thread(() -> System.out.println("   [order varies] A says hi"));
Thread b = new Thread(() -> System.out.println("   [order varies] B says hi"));
a.start(); b.start(); a.join(); b.join();

Thread daemon = new Thread(() -> { while (true) Thread.onSpinWait(); }, "background-spinner");
daemon.setDaemon(true);                                     // before start()
daemon.start();
```

Output:

```
main thread is called 'main'
1 subclass  running in greeter
2 Runnable  running in worker-2
3 lambda    running in worker-3
4 builder   running in worker-4
run() called directly executes in main
state after run(): NEW
start() twice -> IllegalThreadStateException
   [order varies] A says hi
   [order varies] B says hi
both finished (join guarantees this line comes last)
daemon? true, alive? true, id>0? true
main ends; the JVM exits even though the daemon still spins
```

What to notice:

- Calling `run()` **directly** does *not* create a thread: the code runs in `main`, and the `Thread` object stays `NEW`.
- A and B were started together, so either may print first — that's what `[order varies]` means. `join()` guarantees the line after it runs only when both are done.
- **Daemon threads** are background helpers (e.g. the garbage collector's threads). The JVM exits when only daemon threads remain — so the infinite spinner didn't keep the program alive. Normal (non-daemon) threads *do* keep the JVM running.

Useful `Thread` methods

Method	Purpose
`Thread.currentThread()`	the thread running this code
`getName()` / `setName()`	names appear in logs and thread dumps — always name your threads
`threadId()` (19)	unique id (`getId()` is deprecated)
`Thread.sleep(ms)` / `sleep(Duration)` (19)	pause the current thread; throws `InterruptedException`
`join()` / `join(ms)` / `join(Duration)` (19)	wait for another thread to finish
`isAlive()`, `getState()`	inspect
`setDaemon(true)`	background thread (set before `start`)
`setPriority(n)`	a *hint* (1–10) the OS may ignore — never rely on it for correctness
`setUncaughtExceptionHandler(h)`	what to do if `run()` throws (default: print stack trace; the thread dies, others continue)
`Thread.onSpinWait()` (9)	hint inside busy-wait loops
`Thread.yield()`	hint to let others run; rarely useful

Trade-offs: `extends Thread` vs `Runnable`

`extends Thread`	`implements Runnable` / lambda
Uses up your single superclass	Class can extend something else
Couples *what* to do with *how* it runs	The same task can run in a `Thread`, a pool, a virtual thread
Rarely the right choice	Preferred

When not to use raw threads

In application code you should almost never call `new Thread(...)` directly: creating a platform thread is expensive (an OS thread with a 1–2 MB stack reservation by default, depending on the platform), and unmanaged threads are hard to shut down and monitor. Use an `ExecutorService` (§14.11) or, for I/O-heavy tasks on Java 21+, virtual threads (Chapter 15).

Self-check

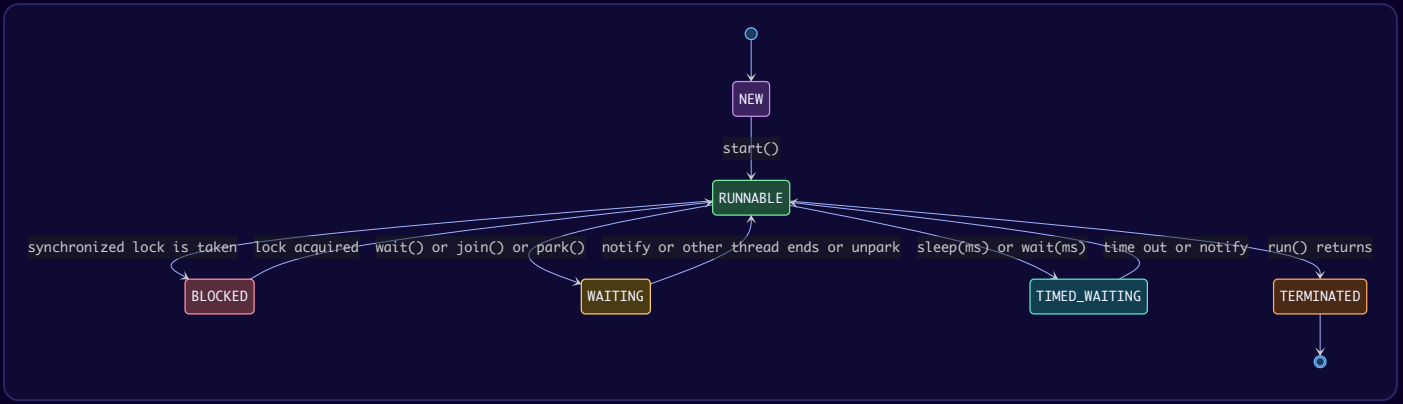
1. What's the difference between `t.start()` and `t.run()`? → `start` creates a new thread that calls `run`; calling `run` directly executes on the current thread.
2. Can you restart a finished thread? → No: `IllegalThreadStateException`. Create a new `Thread` (or reuse the `Runnable`).
3. Why doesn't a running daemon thread stop the JVM from exiting? → The JVM exits when all non-daemon threads have finished.

14.3 Thread lifecycle

What it is

`Thread.getState()` returns one of six `Thread.State` values:

State	Meaning	Typical cause
`NEW`	created, not yet started	`new Thread(...)`
`RUNNABLE`	running **or ready to run** (waiting for a CPU, or even blocked in OS-level I/O)	after `start()`
`BLOCKED`	waiting to **acquire a monitor lock** (`synchronized`) held by another thread	entering a `synchronized` block that's taken
`WAITING`	waiting indefinitely for another thread's action	`Object.wait()`, `join()`, `LockSupport.park()` (used by `Lock`, latches, queues)
`TIMED_WAITING`	waiting with a timeout	`sleep(ms)`, `wait(ms)`, `join(ms)`, `tryLock(timeout)`
`TERMINATED`	`run()` has finished (normally or by exception)	end of `run()`



`code/ch14/threads/Lifecycle.java` drives one thread through every state, polling `getState()` until each is reached (so the output is deterministic):

```
Thread t = new Thread(() -> {
    while (keepSpinning) Thread.onSpinWait();   // RUNNABLE
    Thread.sleep(300);                          // TIMED_WAITING
    go.await();                                 // WAITING (a CountDownLatch parks the thread)
    synchronized (LOCK) { }                     // BLOCKED while main holds LOCK
});
```

Output:

```
after new:              NEW
while spinning:         RUNNABLE
during sleep(300):      TIMED_WAITING
during latch.await():   WAITING
waiting for a monitor:  BLOCKED
after run() returns:    TERMINATED
```

Note the subtle split: waiting for a `synchronized` monitor is `BLOCKED`, but waiting for a `ReentrantLock` is `WAITING` (locks park the thread). A thread blocked reading a socket shows as `RUNNABLE` — the JVM doesn't distinguish OS-level I/O waits. Thread dumps (`jstack <pid>`, `jcmd <pid> Thread.print`) show these states and are the first tool for diagnosing a hung program. More: LLD §12.3.

14.4 Interruption: cooperative cancellation

What it is

Java has no safe way to forcibly kill a thread (`Thread.stop()` was deprecated because it could leave objects half-updated; since Java 20 it throws `UnsupportedOperationException`, and JDK 26 removed it entirely). Instead, one thread **requests** another to stop by calling `t.interrupt()`, which sets the target's **interrupt flag**. The target decides how and when to stop. Analogy: tapping a colleague on the shoulder, not pulling their chair away.

How it works

1. `t.interrupt()` sets `t`'s interrupt flag.
2. If `t` is blocked in an **interruptible** method (`sleep`, `wait`, `join`, `BlockingQueue.take`, `Lock.lockInterruptibly`, `Future.get` …), that method wakes up and throws `InterruptedException` – and **clears the flag**.
3. If `t` is busy computing, nothing happens automatically; the code must check `Thread.currentThread().isInterrupted()` periodically.
4. `Thread.interrupted()` (static) returns the current thread's flag **and clears it**; `isInterrupted()` only reads it.

`code/ch14/threads/Interruption.java` (key parts):

```
try {
    Thread.sleep(60_000);
} catch (InterruptedException e) {
    // flag was cleared when the exception was thrown
    Thread.currentThread().interrupt();          // restore it so callers can see the request
}

while (!Thread.currentThread().isInterrupted()) loops++;     // busy code polls the flag

Thread.currentThread().interrupt();
Thread.interrupted();   // true  (and clears)
Thread.interrupted();   // false
```

Output:

```
sleeper: woke up early, flag now = false
sleeper: flag restored = true
worker: saw the interrupt and stopped cleanly (loops > 0: true)
interrupted() first call  = true
interrupted() second call = false
```

Common mistakes

- **Swallowing** `InterruptedException`: `catch (InterruptedException e) {}`. The cancellation request is lost; `shutdownNow()` on a pool then can't stop the task. Either **rethrow** it (declare `throws InterruptedException`) or **restore the flag** with `Thread.currentThread().interrupt()` – JCIP's rule, and why every `catch` in this chapter's code does so.
- Using a plain `boolean` field as a stop flag without `volatile` (§14.7). Interruption is preferable because it also wakes blocked threads.
- Assuming interruption stops *non-interruptible* blocking such as classic `java.io` socket reads – closing the socket is the way to unblock those.

In the interview

30-second answer: "Interruption is cooperative cancellation: `interrupt()` sets a flag; blocking methods respond by throwing `InterruptedException` and clearing the flag; long computations should poll

> `isInterrupted()`. When I catch `InterruptedException` and can't rethrow it, I restore the flag with `Thread.currentThread().interrupt()`."

- *Why was `Thread.stop` removed?* → It released all locks mid-operation, exposing corrupted state.

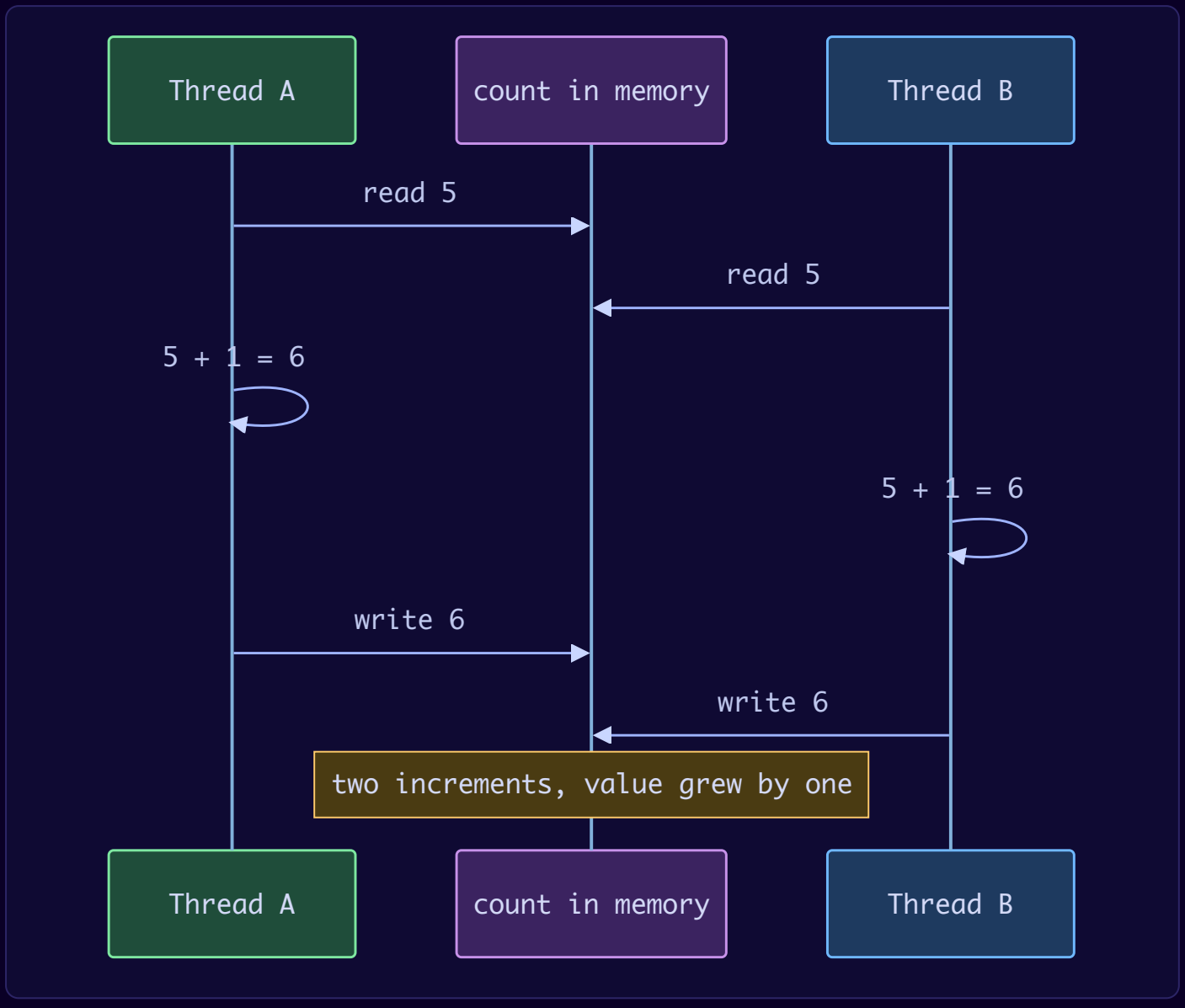
14.5 Race conditions

What it is

A **race condition** is a bug where the result depends on the unlucky timing (interleaving) of threads. The two classic shapes:

1. **Read-modify-write**: `count++` is really *read* `count` → *add* 1 → *write* back. If two threads both read 5, both write 6: one increment is **lost**.
2. **Check-then-act**: `if (seat.isFree()) seat.book(me);` — between the check and the act another thread can book the same seat. Lazy initialisation (`if (instance == null) instance = new X();`) has the same shape.

A **data race**, more narrowly, is when two threads access the same variable without synchronisation and at least one writes. Data races also cause *visibility* problems (§14.7).



`code/ch14/race/RaceCondition.java` (key parts) — 4 threads each add 1,000,000:

```
run(() -> { for (int i = 0; i < PER_THREAD; i++) unsafeCount++; });
run(() -> { for (int i = 0; i < PER_THREAD; i++) synchronized (LOCK) { syncCount++; } });
run(() -> { for (int i = 0; i < PER_THREAD; i++) atomicCount.incrementAndGet(); });
run(() -> { for (int i = 0; i < PER_THREAD; i++) adder.increment(); });

void bookUnsafe(String who) {
    if (owner == null) {          // check ...
        sleepQuietly();           // (real work would widen the window like this)
        owner = who;              // ... then act
        successes++;
    }
}
synchronized void bookSafe(String who) { ...same body... }
```

Output (two runs; the first line differs every time):

```
unsynchronized count++ : 1,127,175   [nondeterministic - lost updates, changes every run]
synchronized           : 4,000,000
AtomicInteger          : 4,000,000
LongAdder              : 4,000,000
unsafe booking attempt 1: successful bookings = 2   [nondeterministic - 2 means double-booked]
unsafe booking attempt 2: successful bookings = 2   [nondeterministic - 2 means double-booked]
unsafe booking attempt 3: successful bookings = 2   [nondeterministic - 2 means double-booked]
synchronized booking: successful bookings = 1
```

```
unsynchronized count++ : 1,927,211   [nondeterministic - lost updates, changes every run]
...
```

Almost **three quarters of the increments vanished** in the first run, and a different number in the second. The seat was double-booked every time. Both fixed versions are exact.

Why these bugs are nasty

- They may **never appear** in tests on a laptop and then appear under production load.
- Adding a `println` to debug them changes the timing and often hides them ("Heisenbugs").
- The JIT compiler may keep a non-volatile field in a register for a whole loop, making results even stranger than simple interleaving suggests.

How to fix (preview of the rest of the chapter)

1. **Don't share** – confine data to one thread (local variables, `ThreadLocal`) – §14.14.
2. **Don't mutate** – share only immutable objects (records with immutable fields) – §14.14.
3. **Synchronise** every access to shared mutable state – `synchronized` (§14.6), locks (§14.9).
4. **Use atomic or concurrent building blocks** – `AtomicInteger`, `ConcurrentHashMap.merge` (§14.10, §14.12).

More examples and the "critical section" framing: LLD §12.5.

Self-check

1. Why is `count++` not atomic? → It's three steps: read, add, write.
2. Name the other classic race shape. → Check-then-act (e.g. lazy initialisation, "if free then book").
3. Does making a field `volatile` fix `count++`? → No (§14.7).

14.6 `synchronized` and intrinsic locks

What it is

Every Java object has a built-in lock called its **intrinsic lock** or **monitor**. A `synchronized` block or method acquires it on entry and releases it on exit – even if an exception is thrown. While one thread holds a

monitor, other threads trying to enter a block synchronised on the *same object* wait (state `BLOCKED`).

`synchronized` gives two guarantees:

1. **Mutual exclusion (atomicity)**: only one thread at a time runs code guarded by that lock.
2. **Visibility**: everything a thread wrote before releasing a lock is visible to the next thread that acquires the *same* lock (a *happens-before* edge — Chapter 15 §15.8).

How it works

Form	Lock used
`synchronized void m()` (instance method)	`this`
`static synchronized void m()`	the `Class` object (`Counter.class`)
`synchronized (obj) { ... }`	`obj` — any object you choose

Intrinsic locks are **reentrant**: a thread that holds a lock can enter another block guarded by the same lock (e.g. a synchronized method calling another synchronized method on `this`) without deadlocking itself. The JVM counts the entries.

`code/ch14/sync/SynchronizedBasics.java` (key parts):

```
static class Counter {
    private int value;
    public synchronized void increment() { value++; }                 // locks this
    public void incrementBlock() { synchronized (this) { value++; } }  // same lock
    public synchronized int get() { return value; }                   // reads need it too
    public synchronized void addTwice() { increment(); increment(); }  // reentrant
    public static synchronized String describe() { ... }              // locks Counter.class
}
synchronized (c) { Thread.holdsLock(c); Thread.holdsLock(Counter.class); }

Object lockA = new Object(), lockB = new Object();       // different locks don't exclude each other
```

Output:

```
4 threads x 500,000 synchronized increments = 2000000
reentrant call worked, value = 2000002
static synchronized locks Counter.class
inside synchronized(c): holdsLock(c)=true holdsLock(Counter.class)=false
different locks -> both threads were inside at once (latch reached 0: true)
```

The last line proves the most common mistake: protection only works if **all** code touching the data uses **the same lock**. Here the two threads each held a different lock, so both were "inside" at once (the latch could only reach 0 because both got in).

Guidelines

- **Guard every access** — reads too. An unsynchronised `get()` might return a stale value (visibility).
- **Keep critical sections short**; never do slow I/O or call unknown code (callbacks, listeners) while holding a lock — that's how deadlocks and stalls happen.
- **Prefer a private final lock object** (`private final Object lock = new Object();`) over `synchronized(this)` in library code, so outside code can't grab your lock and interfere.
- **Never lock on** `String` literals, boxed integers (`Integer` cache!) or other shared/interned objects, and not on a field that can be reassigned.
- Document which lock guards which fields (JCIP's `@GuardedBy("lock")` convention).
- Since Java 24 (JEP 491), `synchronized` no longer "pins" virtual threads — details in Chapter 15 §15.5.

Common mistakes

- Synchronising the writer but not the reader.

- Synchronising two methods on *different* objects and assuming they're mutually exclusive (static vs instance synchronized methods use different locks!).
- Thinking `Collections.synchronizedList(list)` makes *iteration* safe — you must still `synchronized (list) { for (...) }` (§14.12).

In the interview

> **30-second answer:** "`synchronized` acquires the monitor of an object — `this` for instance methods, the `Class` for static ones, or any object for a block. It gives mutual exclusion and visibility, it's reentrant, and the lock is released automatically, even on exceptions. All accesses to a piece of shared state must use the same lock."

- *Is `synchronized` reentrant?* → Yes; the same thread can re-acquire a lock it holds.
- *Static and instance synchronized methods — do they block each other?* → No; different locks (`Class` object vs instance).

Deeper treatment with diagrams: LLD §12.7.

14.7 `volatile` and visibility

What it is

Modern CPUs keep copies of memory in per-core caches, and both the compiler and the CPU reorder instructions for speed. So without synchronisation, a value written by one thread **may never become visible** to another thread, or become visible out of order. A field declared `volatile` fixes visibility:

- Every read of a volatile field sees the **most recent write** by any thread.
- Writes made *before* a volatile write become visible to a thread that later *reads* that volatile field (it creates a happens-before edge — Chapter 15 §15.8).
- `volatile long` / `double` reads and writes are atomic (plain 64-bit writes are allowed to be split in two by the JLS).

`volatile` does **not** provide mutual exclusion: `volatileCounter++` is still read-modify-write.

`code/ch14/sync/VolatileFlag.java` (key parts):

```
private static volatile boolean running = true;          // without volatile the worker MAY spin forever

Thread worker = new Thread(() -> { long spins = 0; while (running) spins++; ... });
worker.start();
Thread.sleep(200);
running = false;                                         // guaranteed to be seen

// 4 threads x 1,000,000 of volatileCounter++
```

Output:

```
worker saw running=false and stopped (spins > 0: true)
volatile count++ from 4 threads: 1,113,395 of 4,000,000   [nondeterministic - still loses updates]
```

(Why "may spin forever" without `volatile`: the JIT is allowed to read `running` once and turn the loop into `while (true)`. Whether it does depends on the JVM and timing, which is exactly why you can't test your way to safety — follow the rules instead.)

When to use `volatile`

- A **status flag** written by one thread and read by others (`shutdownRequested`).
- **Publishing** an immutable object reference (`volatile Config current;` replaced wholesale).

- The double-checked-locking idiom (Chapter 15 §15.8).

When not to use it

When the new value depends on the old (`count++`, `balance -= x`), or several fields must change together — use locks or atomics.

	`volatile`	`synchronized`	`AtomicInteger`
Visibility	✓	✓	✓
Atomic compound update (`++`)	✗	✓	✓ (single variable)
Multi-variable invariants	✗	✓	✗
Blocking	never	may block	never (retries)

More: LLD §12.6.

14.8 `wait` / `notify`: guarded blocks

What it is

Sometimes a thread must wait until a *condition* becomes true ("the queue is not empty"). Every object's monitor has a **wait set**:

- `obj.wait()` — releases `obj`'s lock and sleeps until notified (or interrupted, or a *spurious wakeup*); then re-acquires the lock before returning.
- `obj.notify()` — wakes **one** waiting thread (which one is unspecified). `obj.notifyAll()` wakes **all** of them.
- All three must be called **while holding `obj`'s lock**, or you get `IllegalMonitorStateException`.

How it works — the guarded-block idiom

1. Acquire the lock (`synchronized`).
2. `while` the condition is false, `wait()`. (A `while`, not `if`: after waking, the condition may already be false again — another thread got there first — or the wakeup may be spurious.)
3. Do the action.
4. After changing state that others wait on, `notifyAll()`.

`code/ch14/sync/WaitNotify.java` — a two-slot buffer between a producer and a consumer:

```
public synchronized void put(T item) throws InterruptedException {
    while (items.size() == capacity) wait();     // wait for "not full"
    items.add(item);
    notifyAll();                                 // wake consumers
}
public synchronized T take() throws InterruptedException {
    while (items.isEmpty()) wait();              // wait for "not empty"
    T item = items.remove();
    notifyAll();                                 // wake producers
    return item;
}
```

Output:

```
consumer received in order: 1 2 3 4 5 6 7 8 9 10
sum = 55 (1..10 = 55)
wait() without the lock -> IllegalMonitorStateException
```

The producer ends with a **poison pill** (`-1`), a special value meaning "no more items" — a common way to stop consumers.

When not to use it

In real code, use `BlockingQueue` (§14.12), `CountDownLatch` / `Semaphore` (§14.13) or `Condition` (§14.9), which are easier to get right. Know `wait` / `notify` for interviews and legacy code. `notify()` (single) is only safe when all waiters wait for the same condition and any one can proceed; otherwise a "lost wakeup" can leave threads stuck – `notifyAll()` is the safe default. Step-by-step coverage: LLD §12.13 and the blocking queue built two ways in LLD §13.8.

Self-check

1. Why `while` and not `if` around `wait()`? → Spurious wakeups, and the condition may have changed before the thread re-acquired the lock.
2. Does `wait()` release the lock? → Yes (only that object's lock); `sleep()` does not release any lock.

14.9 Explicit locks: `ReentrantLock`, `Condition`, `ReadWriteLock`, `StampedLock`

What it is

`java.util.concurrent.locks` (Java 5) provides lock *objects* with the same mutual-exclusion and visibility guarantees as `synchronized`, plus features `synchronized` lacks:

Feature	`synchronized`	`ReentrantLock`
Released automatically	✓ (end of block)	✗ – must `unlock()` in `finally`
Try without waiting / with timeout	✗	`tryLock()`, `tryLock(time, unit)`
Interruptible wait for the lock	✗	`lockInterruptibly()`
Fairness (FIFO-ish order)	✗	`new ReentrantLock(true)`
Multiple wait conditions	one wait set per object	many `Condition`s per lock
Lock/unlock in different methods	✗	✓ (use with care)
Inspect (`isLocked`, `getHoldCount`, queued threads)	✗	✓
Syntax	concise, can't forget to unlock	verbose

How it works

```
lock.lock();
try {
    // critical section
} finally {
    lock.unlock();          // ALWAYS in finally, or an exception leaves it locked forever
}
```

`code/ch14/locks/LocksDemo.java` (key parts):

```
lock.lock(); try { lock.lock(); try { ... } finally { lock.unlock(); } } finally { lock.unlock(); }  // reentrant
boolean got = lock.tryLock(150, TimeUnit.MILLISECONDS);      // another thread holds it
lock.lockInterruptibly();                                     // in a waiter thread, then waiter.interrupt()

Condition ready = l.newCondition();
while (!flag[0]) ready.await();                               // like wait(); signalAll() like notifyAll()

ReadWriteLock rw = new ReentrantReadWriteLock();
rw.readLock().lock();   // 3 readers wait at a CyclicBarrier(3) while holding the read lock
rw.writeLock().lock();  // exclusive: a reader's tryLock fails
```

Output:

```
holding lock: isHeldByCurrentThread=true holdCount=1
re-entered: holdCount=2
tryLock(150ms) while another thread holds it -> false after >=150ms: true
waiter: interrupted while waiting for the lock
Condition: woke up, flag = true
ReadWriteLock: readers holding the read lock simultaneously = 3
writer holds write lock; can a reader get in now? false
```

The barrier trick proves read-sharing: a `CyclicBarrier(3)` only opens when three threads are waiting at it, and they were all holding the read lock at the time.

`ReadWriteLock` and `StampedLock`

- `ReentrantReadWriteLock`: many readers **or** one writer. Worth it when reads vastly outnumber writes and critical sections are not tiny (e.g. an in-memory catalogue refreshed hourly). For small, fast sections a plain lock or `ConcurrentHashMap` is usually faster.
- `StampedLock` (Java 8): adds **optimistic reads** – read without locking, then `validate(stamp)` to check that no write happened; fall back to a real read lock if one did. Very fast for read-mostly data, but **not reentrant** and doesn't support `Condition`s – for experts. Examples: LLD §12.10.

When not to use explicit locks

If you don't need `tryLock`, timeouts, interruptibility, fairness or several conditions, **prefer** `synchronized`: shorter, impossible to forget to unlock, and (since Java 24) equally friendly to virtual threads. Fair locks have much lower throughput; use them only when starvation is a real problem.

In the interview

> **30-second answer:** "`ReentrantLock` offers what `synchronized` doesn't: `tryLock` with timeouts to avoid deadlock, interruptible acquisition, optional fairness and multiple `Condition`s. The price is that I must unlock in `finally`. I default to `synchronized` and switch to `ReentrantLock` when I need those features; `ReadWriteLock` for read-heavy data, `StampedLock` for optimistic reads."

- *How does `tryLock` help against deadlock?* → A thread that can't get the second lock backs off, releases the first and retries, breaking the circular wait (Chapter 15 §15.9).

14.10 Atomic variables

What it is

`java.util.concurrent.atomic` classes (`AtomicInteger`, `AtomicLong`, `AtomicBoolean`, `AtomicReference<V>`, `AtomicIntegerArray`, …) offer single-variable operations that are **atomic without locks**. They're built on the CPU's **compare-and-set (CAS)** instruction: "if the value is still `expected`, set it to `new` and return true;

otherwise change nothing and return false". A thread that loses the race simply **retries**; no thread ever blocks.

Analogy: editing a shared document with version numbers — "save my change only if the document is still version 17; otherwise reload and try again."

`code/ch14/atomic/AtomicDemo.java` (key parts):

```
AtomicInteger n = new AtomicInteger(10);
n.incrementAndGet();  n.getAndIncrement();  n.addAndGet(5);
n.compareAndSet(17, 100);                  // true: value was 17
n.compareAndSet(17, 200);                  // false: value is 100 now
n.updateAndGet(x -> x * 2);
n.accumulateAndGet(50, Math::max);

do { prev = hits.get(); next = prev + 1; } while (!hits.compareAndSet(prev, next));   // what updateAndGet does

AtomicReference<Range> range = new AtomicReference<>(new Range(0, 10));
range.updateAndGet(r -> new Range(r.low() + 1, r.high() + 1));     // 4 threads x 10,000

LongAdder requests = new LongAdder();       // 8 threads x 100,000 increments
```

Output:

```
incrementAndGet -> 11
getAndIncrement -> 11 (now 12)
addAndGet(5)    -> 17
CAS 17->100     -> true, value 100
CAS 17->200     -> false, value 100 (expected value was stale)
updateAndGet x2 -> 200
accumulateAndGet(max with 50) -> 200
hand-written CAS loop -> 1
AtomicReference after 40,000 shifts -> Range[low=40000, high=40010]
LongAdder sum -> 800000
```

- `AtomicReference` + an **immutable** record lets you update several related values (low and high) atomically: build a new object, CAS it in. The invariant `high - low == 10` always holds.
- The function passed to `updateAndGet` may run **more than once** (on retries), so it must have no side effects.
- `LongAdder` (Java 8) spreads increments over several internal cells to avoid CAS contention; `sum()` adds them up. Much faster than `AtomicLong` for hot counters (metrics, request counts), but `sum()` is not an atomic snapshot and there is no `compareAndSet`.

Trade-offs

	Locks	Atomics
Scope	any number of variables	one variable (or one reference to an immutable object)
Under low contention	slower	very fast
Under very high contention	threads sleep	many CAS retries burn CPU (`LongAdder` helps)
Deadlock possible?	yes	no

CAS internals, the ABA problem and `AtomicStampedReference`: LLD §12.15.

Self-check

1. What does `compareAndSet` return if the current value isn't the expected one? → `false`, and the value is unchanged.
2. When is `LongAdder` better than `AtomicLong`? → Many threads updating, rare reads.
3. Why must the `updateAndGet` lambda be side-effect free? → It may be retried.

14.11 Executors and thread pools

What it is

An `ExecutorService` separates **submitting tasks** from **running them**. You hand it `Runnable`s or `Callable`s; it runs them on a **thread pool** – a set of reusable worker threads fed from a **work queue**. Analogy: a taxi rank. Passengers (tasks) queue; a fixed number of taxis (threads) take the next passenger when free, instead of buying a new car for every trip.

- `Runnable` – `void run()`, can't return a value or throw checked exceptions.
- `Callable<V>` – `V call() throws Exception`, returns a result.
- `Future<V>` – a handle to a pending result: `get()` (blocks), `get(timeout, unit)`, `isDone()`, `cancel(mayInterrupt)`, `isCancelled()`, and since Java 19 `state()`, `resultNow()`, `exceptionNow()`.

Why it matters

1. **Reuse**: creating an OS thread per task costs time and memory; pools amortise it.
2. **Bounded resources**: a pool of 20 threads can't create 10,000 threads during a traffic spike.
3. **Lifecycle management**: one place to shut everything down, wait for completion, collect results and exceptions.

How it works

1. Create a pool (factory methods below, or a `ThreadPoolExecutor` directly).
2. `execute(runnable)` (fire and forget) or `submit(callable)` → `Future`.
3. `invokeAll(tasks)` → all `Future`s when every task is done; `invokeAny(tasks)` → the first successful result, cancelling the rest.
4. Shut down: `shutdown()` (stop accepting, finish queued), `awaitTermination(timeout)` (wait), `shutdownNow()` (interrupt running tasks, return queued ones). Since **Java 19** `ExecutorService` is `AutoCloseable`: **`close()`** = shutdown + wait, so try-with-resources works.

`code/ch14/executors/ExecutorBasics.java` (key parts):

```
try (ExecutorService pool = Executors.newFixedThreadPool(3)) {
    pool.execute(() -> ...);                                        // 6 tasks
    Future<Integer> f = pool.submit(() -> { Thread.sleep(100); return 6 * 7; });
    f.isDone(); f.get(); f.state(); f.resultNow();

    Future<Integer> bad = pool.submit(() -> Integer.parseInt("forty-two"));
    bad.get();                                                      // ExecutionException wrapping the cause

    Future<String> slow = pool.submit(() -> { Thread.sleep(10_000); return "late"; });
    slow.get(100, TimeUnit.MILLISECONDS);                           // TimeoutException
    slow.cancel(true);                                              // interrupts the sleeping task

    pool.invokeAll(List.of(() -> "a", () -> { Thread.sleep(50); return "b"; }, () -> "c"));
    pool.invokeAny(List.of(() -> { Thread.sleep(300); return "slow mirror"; },
                           () -> { Thread.sleep(10);  return "fast mirror"; }));
}                                                                   // close(): waits for all tasks

ex.shutdown(); ex.awaitTermination(5, TimeUnit.SECONDS); ex.submit(...);   // rejected

try (ScheduledExecutorService sched = Executors.newScheduledThreadPool(1)) {
    sched.scheduleAtFixedRate(() -> ..., 50, 100, TimeUnit.MILLISECONDS);
    sched.schedule(() -> "delayed hello", 50, TimeUnit.MILLISECONDS).get();
}
```

Output:

```
6 tasks ran on at most 3 threads: true
isDone right after submit? false
get() blocks, then returns 42
state = SUCCESS, resultNow = 42
get() -> ExecutionException caused by NumberFormatException
get(100ms) -> TimeoutException; cancel(true) = true, isCancelled = true
invokeAll results in order: abc
invokeAny -> fast mirror
pool closed (all tasks finished)
single-thread executor task
awaitTermination -> true, isTerminated = true
submit after shutdown -> RejectedExecutionException
tick 1 at ~50 ms   [timing approximate]
tick 2 at ~150 ms   [timing approximate]
tick 3 at ~250 ms   [timing approximate]
delayed hello
```

- An exception inside a task **doesn't crash anything**: it's stored in the `Future` and rethrown by `get()` as `ExecutionException` with the original as its *cause*. If you never call `get()`, a `submit`ted task's exception is silently lost — a frequent source of "the job just stopped working" mysteries. (`execute()`d tasks, by contrast, let the exception reach the thread's uncaught-exception handler, which prints it.)
- `invokeAll` returns results in **submission order**, not completion order.
- `scheduleAtFixedRate` runs at a fixed *rate* (start times 100 ms apart); `scheduleWithFixedDelay` waits a fixed *delay* after each run ends. If a periodic task throws, its future runs are silently cancelled.

The `Executors` factory methods

Factory	Pool	Queue	Watch out
`newFixedThreadPool(n)`	exactly n threads	**unbounded** `LinkedBlockingQueue`	a flood of tasks piles up in memory
`newCachedThreadPool()`	0…∞ threads, idle ones die after 60 s	hand-off (`SynchronousQueue`)	a flood creates unbounded threads
`newSingleThreadExecutor()`	1 thread	unbounded	tasks run sequentially in order
`newScheduledThreadPool(n)`	n threads	delay queue	periodic tasks die silently on exception
`newWorkStealingPool()` (8)	a `ForkJoinPool`	per-thread deques	for many small CPU tasks
`newVirtualThreadPerTaskExecutor()` (21)	a new virtual thread per task	none	not a pool at all — Chapter 15

Configuring `ThreadPoolExecutor`

For production services, build the pool explicitly so every limit is a decision. The rules for a new task are:

1. Fewer than **core** threads running → start a new thread.
2. Otherwise → put it in the **queue**.
3. Queue full and fewer than **max** threads → start an extra thread.
4. Otherwise → call the **rejection policy**.

`code/ch14/executors/PoolConfig.java` (key parts) — core 2, max 3, queue capacity 2, 7 tasks that all block:

```
ThreadPoolExecutor pool = new ThreadPoolExecutor(
        2, 3, 30, TimeUnit.SECONDS,
        new ArrayBlockingQueue<>(2),
        new ThreadPoolExecutor.AbortPolicy());
```

Output:

```
available processors: 14   [machine-dependent]
task 1 accepted  (threads=1, queued=0)
task 2 accepted  (threads=2, queued=0)
task 3 accepted  (threads=2, queued=1)
task 4 accepted  (threads=2, queued=2)
task 5 accepted  (threads=3, queued=2)
task 6 REJECTED  (threads=3, queued=2)
task 7 REJECTED  (threads=3, queued=2)
completed tasks = 5
third task ran in: main (the caller)
```

Note the surprise in step 3: the pool grows beyond the core size only when the queue is **full** – so with an unbounded queue, `max` is never used.

Rejection policies: `AbortPolicy` (default, throws `RejectedExecutionException`), `CallerRunsPolicy` (the submitting thread runs the task itself – natural back-pressure that slows the producer; shown in the last line), `DiscardPolicy` (silently drop), `DiscardOldestPolicy` (drop the oldest queued task).

Sizing (JCIP): for CPU-bound work, about N_{cpu} **(+1)** threads – more just adds context switches. For tasks that wait, $N_{threads} \approx N_{cpu} \times$ **(1 + wait time / compute time)**; e.g. 8 cores, tasks that wait 90 ms per 10 ms of CPU → ~80 threads. Then measure. For I/O-bound tasks on Java 21+, virtual threads usually remove the need for this calculation.

Common mistakes

- Forgetting to shut down a pool → non-daemon worker threads keep the JVM alive forever.
- Unbounded queues/pools in servers → `OutOfMemoryError` under load instead of a clean rejection.
- Tasks that wait on other tasks **in the same bounded pool** → **thread-starvation deadlock** when all threads wait for queued tasks that can't run.
- Ignoring `Future`s from `submit` → exceptions disappear.
- Sharing one pool for fast and slow work → slow tasks hog all threads (use separate pools, the *bulkhead* pattern).

In the interview

> **30-second answer:** "I use an `ExecutorService` rather than raw threads: tasks go in as `Runnable` / `Callable`, results and exceptions come back through `Future`, and I shut it down with `shutdown` / `awaitTermination` or try-with-resources on Java 19+. In production I configure `ThreadPoolExecutor` explicitly – bounded queue, a sized max, a rejection policy like `CallerRunsPolicy` for back-pressure, and named threads."

- *`shutdown` vs `shutdownNow`?* → Graceful (queued tasks still run) vs interrupt running tasks and return the queued ones.
- *Why is `newFixedThreadPool` risky?* → Its queue is unbounded.
- *`submit` vs `execute`?* → `submit` returns a `Future` and captures exceptions; `execute` returns nothing and exceptions go to the uncaught handler.

Design discussion: LLD §12.21 Thread pool pattern. Exam tables (`TimeUnit`, `ScheduledExecutorService` methods): OCP Chapter 13.

14.12 Concurrent collections

What it is

The ordinary collections of Chapter 10 (`ArrayList`, `HashMap`) are not thread-safe. There are two ways to share collections:

1. **Synchronised wrappers** – `Collections.synchronizedList/Map/Set(...)`, and the legacy `Vector` / `Hashtable`: every method takes one lock. Simple, but only one thread at a time, and **compound actions and iteration**

still need external locking.

2. **Concurrent collections** (`java.util.concurrent`) — designed for concurrent access with fine-grained locking or lock-free algorithms, plus **atomic compound operations** built in.

Class	Use for	Notes
`ConcurrentHashMap`	shared maps, caches, counters	atomic `putIfAbsent`, `computeIfAbsent`, `compute`, `merge`; reads don't block; **no null keys/values**; weakly consistent iterators
`ConcurrentSkipListMap` / `Set`	sorted concurrent map/set	concurrent `TreeMap`; O(log n)
`CopyOnWriteArrayList` / `Set`	read-mostly lists (listeners, config)	every write copies the array; iterators see a snapshot, never throw CME
`ArrayBlockingQueue`	bounded producer-consumer	`put` waits when full, `take` waits when empty
`LinkedBlockingQueue`	producer-consumer (optionally bounded)	unbounded by default
`PriorityBlockingQueue`, `DelayQueue`	priority / delayed tasks	unbounded
`SynchronousQueue`	direct hand-off (capacity 0)	used by cached thread pools
`LinkedTransferQueue`	producer can wait until consumed	
`ConcurrentLinkedQueue` / `Deque`	non-blocking queue	lock-free; `poll` returns `null` when empty

`BlockingQueue` methods come in four flavours: throw (`add` / `remove`), special value (`offer` / `poll` → `false` / `null`), **block** (`put` / `take`), and **time out** (`offer(e, t, unit)` / `poll(t, unit)`).

`code/ch14/collections/ConcurrentCollections.java` (key parts):

```
counts.merge(w, 1, Integer::sum);                                   // atomic per key

Integer old = syncMap.get(w);                                       // each call is thread-safe,
syncMap.put(w, old == null ? 1 : old + 1);                          // the combination is NOT

groups.computeIfAbsent(w.substring(0, 1), k -> new CopyOnWriteArrayList<>()).add(w);
counts.put("x", null);                                              // NullPointerException

for (String s : plain) if (s.equals("a")) plain.add("d");           // ArrayList: CME
for (String s : cow)   { seen++; if (s.equals("a")) cow.add("d"); } // snapshot iteration

BlockingQueue<String> queue = new ArrayBlockingQueue<>(3);          // producer put()s, consumer take()s
```

Output:

```
ConcurrentHashMap merge : {and=80000, bat=40000, cat=40000, hat=40000, the=120000}
synchronizedMap get+put : {and=55878, bat=32934, cat=33358, hat=33388, the=76273}   [nondeterministic - lost
updates; expected the=120000]
computeIfAbsent sizes   : {a=8, b=4, c=4, h=4, t=12}
ConcurrentHashMap rejects null values -> NullPointerException
ArrayList modify during for-each -> ConcurrentModificationException
CopyOnWriteArrayList: iterator saw 3 elements, list now [a, b, c, d]
BlockingQueue consumed  : [job-1, job-2, job-3, job-4, job-5]
offer on full queue     : offer(2) = false (returns false instead of blocking)
poll with timeout, empty: null
ConcurrentSkipListMap sorted descending? true
ConcurrentLinkedQueue size after 4 x 1000 offers = 4000
```

The second line is the key lesson: a thread-safe collection does **not** make *your sequence of calls* atomic. `get` then `put` is check-then-act; a third of the counts were lost. Use the atomic methods (`merge`, `compute`, `computeIfAbsent`, `putIfAbsent`) — or hold one lock around the whole sequence.

Common mistakes

- Using `containsKey` + `put` (or `get` + `put`) on a `ConcurrentHashMap` — use `putIfAbsent` / `computeIfAbsent` / `merge`.
- Long or blocking work inside `computeIfAbsent`'s function — it runs while that key's hash bin is locked, so other updates to keys in the same bin wait. Never modify the same map from inside the function.
- `size()` on a busy `ConcurrentHashMap` is an estimate at that moment; don't use it for logic.
- `CopyOnWriteArrayList` for write-heavy lists — each add copies the whole array: O(n).
- Iterating a `synchronizedList` without `synchronized (list) { ... }` → `ConcurrentModificationException` or worse.

In the interview

> **30-second answer:** "For shared maps I use `ConcurrentHashMap` and its atomic methods — `merge`, `computeIfAbsent` — because separate `get` and `put` calls are a check-then-act race even on a thread-safe map. `CopyOnWriteArrayList` for read-mostly lists like listeners, `BlockingQueue` for producer-consumer, and the skip-list classes when I need ordering."

- *How does `ConcurrentHashMap` scale?* → Since Java 8: CAS for inserting into empty bins and a lock on only the first node of a bin for updates; reads are lock-free. (Java 7 used segments.) A simplified build: LLD §13.7.
- *Why no nulls?* → `get` returning `null` must unambiguously mean "absent"; with concurrent updates you can't follow up with `containsKey` safely.

More in LLD §12.16 and §12.22 Producer-consumer.

14.13 Synchronizers: `CountDownLatch`, `CyclicBarrier`, `Semaphore`, `Phaser`

What it is

Synchronizers are objects whose job is to coordinate the *flow* of threads based on their own state.

Synchronizer	Idea	Reusable?	Typical use
`CountDownLatch(n)`	threads `await()` until `countDown()` has been called n times	no (one-shot)	"start all workers together"; "wait until N services are initialised"
`CyclicBarrier(n, action)`	n threads `await()` each other; when the last arrives, the optional action runs and all proceed	yes	iterative algorithms in phases (simulations)
`Semaphore(permits)`	`acquire()` takes a permit (waits if none), `release()` returns one	yes	limit concurrency: at most 10 DB connections / API calls
`Phaser` (7)	like a barrier, but parties can register/deregister dynamically; numbered phases	yes	variable number of participants
`Exchanger<V>`	two threads swap objects at a meeting point	yes	double-buffering (rare)

`code/ch14/synchronizers/Synchronizers.java` (key parts):

```
CountDownLatch startGate = new CountDownLatch(1), finished = new CountDownLatch(n);
// workers: startGate.await(); work...; finished.countDown();
startGate.countDown();  finished.await();

CyclicBarrier barrier = new CyclicBarrier(3, () -> System.out.println("barrier: all 3 arrived, ..."));
// 3 threads: for (round 0..1) barrier.await();

Semaphore permits = new Semaphore(3);
permits.acquire();
try { /* at most 3 threads here */ } finally { permits.release(); }

Phaser phaser = new Phaser(1);  phaser.register();  phaser.arriveAndAwaitAdvance();
```

Output:

```
latch: work before opening the gate = 0
latch: work after all finished = 40, count = 0
barrier: all 3 arrived, phase 1 complete
barrier: all 3 arrived, phase 2 complete
semaphore: 20 tasks on 10 threads, max ever inside = 3 (limit 3)
phaser: main + 2 workers passed phase 0, current phase = 1
```

- `countDown()` goes in `finally` so a failing worker can't leave the waiter stuck forever. Prefer `await(timeout, unit)` in production.
- A `Semaphore` has no owner: any thread can `release()`, and releasing without acquiring *increases* the permits — always pair `acquire` with `release` in `finally`.
- If one thread waiting at a `CyclicBarrier` is interrupted or times out, the barrier becomes **broken** and all others get `BrokenBarrierException`.

Worked examples: LLD §12.12 Semaphores, §12.14 Latch and barrier, and the rate limiter in LLD §13.11.

Self-check

1. Latch or barrier for "main waits until 5 workers finish"? → `CountDownLatch(5)`.
2. How do you cap concurrent calls to an API at 10? → `Semaphore(10)`, acquire/release around each call.
3. Can a `CountDownLatch` be reset? → No; use `CyclicBarrier` or `Phaser` for reuse.

14.14 Designing thread-safe classes

Java Concurrency in Practice boils thread safety down to one sentence: **manage access to shared, mutable state.** You can remove "shared", remove "mutable", or manage the access:

Strategy	How	Example
Thread confinement	only one thread ever touches the data	local variables; objects created and used inside one task; one thread owning a UI; `ThreadLocal` (Chapter 15)
Immutability	no state changes after construction; `final` fields; defensive copies	records with immutable components, `String`, `List.of(...)`, `java.time` – always thread-safe
Encapsulated synchronisation	private state, every access under the same private lock	the `Counter` in §14.6
Delegation	keep state in thread-safe components	a `ConcurrentHashMap` field, an `AtomicLong` counter

Guidelines:

1. **Prefer immutable objects**; make fields `final` unless they must change (Chapter 5).
2. **Don't let `this` escape** from a constructor (starting a thread or registering a listener in the constructor) – other threads could see a half-built object.
3. **Invariants involving several variables** need one lock around all of them (atomics per field aren't enough).
4. **Document** thread-safety guarantees in Javadoc ("This class is thread-safe" / "not thread-safe").
5. **Prefer high-level utilities** (executors, concurrent collections, synchronizers) over `wait`/`notify` and raw threads – *Effective Java* Items 80–81.
6. **Test under load** and use tools: thread dumps, JFR, and stress-testing harnesses such as OpenJDK's jcstress for low-level code – but remember tests can't prove absence of races; reasoning about happens-before can (Chapter 15).

Choosing the tool

Need	Reach for
Run tasks in the background	`ExecutorService` (virtual threads for I/O-bound – Chapter 15)
Counter / accumulator	`AtomicInteger` / `AtomicLong`; `LongAdder` if very hot
Stop flag, publish an immutable snapshot	`volatile`
Multi-field invariant	`synchronized` (or `ReentrantLock` if you need `tryLock`/timeouts)
Shared map / cache	`ConcurrentHashMap` + atomic methods
Producer-consumer	`BlockingQueue`
Wait for N things to finish	`CountDownLatch` (or `invokeAll`, or `CompletableFuture.allOf`)
Limit concurrent access	`Semaphore`
Read-mostly data with rare writes	`CopyOnWriteArrayList`, `ReadWriteLock`, or an immutable snapshot in a `volatile`/`AtomicReference`

The LLD book's version of this table with interview framing: LLD §12.26; how concurrency appears in design interviews (seat booking, parking lots, inventory): LLD §12.25.

In the interview (chapter-level)

- *How would you make a seat-booking method safe?* → Make check-and-book atomic: `synchronized` on the show (or a per-seat lock), or `ConcurrentHashMap.putIfAbsent(seatId, userId)` – and in a real system a database constraint or `SELECT … FOR UPDATE`, because several servers run the code.

- *`synchronized` vs `volatile` vs `AtomicInteger`?* → Mutual exclusion + visibility / visibility only / lock-free atomic single-variable updates.
- *What is a thread-safe class?* → One that stays correct under any interleaving of calls from multiple threads without callers adding synchronisation.

Chapter TL;DR

Topic	Remember
Threads	Share the heap, own their stack; `start()` (not `run()`) creates a thread; `join()` waits; daemon threads don't keep the JVM alive; `Thread.Builder` (21)
States	NEW, RUNNABLE, BLOCKED (monitor), WAITING, TIMED_WAITING, TERMINATED
Interruption	Cooperative; blocking calls throw `InterruptedException` and clear the flag; restore it or rethrow
Races	Read-modify-write and check-then-act; results vary run to run; tests can't prove absence
`synchronized`	Mutual exclusion + visibility; `this` / `Class` / chosen object; reentrant; same lock for every access
`volatile`	Visibility (and ordering) only; perfect for flags; `++` still races
`wait` / `notify`	Hold the lock, `while` loop, prefer `notifyAll`; prefer higher-level tools
Locks	`ReentrantLock`: `tryLock`, timeouts, interruptible, fair, `Condition`s – unlock in `finally`; `ReadWriteLock`; `StampedLock`
Atomics	CAS-based, lock-free, single variable; `AtomicReference` + immutable object; `LongAdder` for hot counters
Executors	Prefer to raw threads; `Future` + `ExecutionException`; shut down (or `close()`, 19); bounded queues + rejection policy in servers; size by CPU vs wait
Collections	`ConcurrentHashMap` atomic methods (`merge`, `computeIfAbsent`); COW lists for read-mostly; `BlockingQueue` for producer-consumer
Synchronizers	Latch (one-shot), barrier (cyclic), semaphore (permits), phaser (dynamic)
Design	Confinement, immutability, encapsulated locking, delegation

Where next: composing asynchronous work with `CompletableFuture`, Fork/Join, **virtual threads**, structured concurrency, scoped values, the **Java Memory Model** and deadlock/livelock/starvation in Chapter 15 – Concurrency II. Practise with classic problems in LLD Chapter 13; exam traps in OCP Chapter 13. Python comparison (GIL, `threading`, `asyncio`): Python Chapter 16.